Vivek Raj

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Experience

Senior Software Engineer September 2026 – Present
Innovapptive *Hyderabad, IN*

Software Development Engineer April 2025 – September 2026
Wavemaker Inc *Hyderabad, IN*

- Led the design and rollout of a production multi-source **RAG + agent platform** powering search and grounded Q&A across **600+** live documentation pages, **50+** Academy video transcripts, **60+** components from Storybook and Marketplace, now the default knowledge layer for internal developer tooling.
- Architected and deployed a Gen UI application leveraging AI agents to dynamically generate data visualizations (tables, charts, lists) based on user prompts, consolidating 1000+ static screens into a single intelligent interface and reducing development overhead by 95%.
- Developed custom MCP (Model Context Protocol) integrations enabling seamless data retrieval through Swagger APIs, allowing the AI agent to construct POST request bodies automatically and eliminating manual data configuration processes.
- Engineered an MCP server for WaveMaker marketplace integration, enabling developers to programmatically add and publish artifacts to the marketplace, streamlining external artifact integration and reducing deployment time by 60%.

Associate Software Engineer June 2023 – April 2025
Wavemaker Inc *Hyderabad, IN*

- Built and maintain comprehensive Storybook documentation for 50+ React Native components, reducing developer onboarding time by 40% and accelerating integration across 15+ client projects.
- Developed an AI-powered chatbot using LLaMA3 and MedLLaMA2 models to streamline symptom analysis, department recommendation, and appointment booking, reducing appointment booking time by 30% and improving symptom analysis accuracy by 25%.
- Conducted in-depth assessment of React Native application performance, identifying 5+ critical limitations and implementing best practices that led to smoother user interface, enhancing performance by 30%.
- Maintained and enhanced the company's marketplace for sharing prefabs, themes, and connectors, ensuring 100% functionality in documentation and code while improving resource accessibility by 40%.
- Created proof of concepts (POCs) for emerging technologies to demonstrate capabilities to new customers, accelerating the onboarding process by 30%.

Web Developer Intern January 2023 – May 2023
Wavemaker Inc *Hyderabad, IN*

- Developed 20+ web pages using HTML, CSS, Bootstrap, and JavaScript, enhancing user interface and experience.
- Architected a full-stack financial management system using HTML, Bootstrap, CSS, JavaScript, and Java, featuring automated expense tracking, bill reminders, and goal setting, improving task efficiency by 40%.
- Implemented theme customization and flexible data export functionality, enhancing user experience and increasing user efficiency by 40%.
- Engineered 34 REST API endpoints using Spring, enhancing application functionality and scalability, contributing to a 25% reduction in response times and improved overall system performance.

Projects

WaveMaker Ecosystem Agent & Docs MCP | *Python, FastAPI, LangGraph, pgvector* Sept 2025 – May 2026

- Built a unified search-and-answer service that queries multiple MCP-backed sources (Docs, Academy, Storybook, Marketplace), fuses results via **Reciprocal Rank Fusion**, and streams grounded, citation-backed answers as NDJSON over FastAPI.
- Implemented the Docs MCP retrieval server on **Postgres + pgvector** with **HNSW** indexing and FTS-based hybrid search, layered with **Jina reranking** (evaluated against self-hosted `bge-reranker-v2-m3`) to lift retrieval precision, exposed as MCP tools (`search_docs`, `get_doc`, `list_doc_types`) with health/readiness probes and connection-pool-bounded concurrency.

- Instrumented end-to-end **Langfuse** tracing for retrieval quality, latency, and token spend; added optional Redis result caching and an LLM **planner** that selects sources and rewrites per-source queries before fusion.

Furnish | *Github: Frontend Backend* March 2023 - April 2023

- Architected a cross-platform e-commerce application using React Native, enabling seamless user experiences on iOS and Android in 2 weeks.
- Integrated 21 RESTful API endpoints using Express.js for secure user authentication, product management, order handling, and payments, improving scalability and reducing response times by 30%.
- Constructed 14 screens and over 13 components, leveraging Formik and Yup for efficient form handling, reducing form validation time by 70% and improving overall user input accuracy.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL (Postgres, MySQL), HTML/CSS
Frameworks: FastAPI, FastMCP, LangGraph, LangChain, React, React Native, Node.js, Express.js, Spring
AI / LLM: RAG (hybrid search, reranking), AI Agents, MCP (Model Context Protocol), LiteLLM, OpenAI, Langfuse
Data & Retrieval: PostgreSQL, pgvector (HNSW), Full-Text Search, Vector DBs, MongoDB, Firebase Firestore
Developer Tools: Git, Docker, Jenkins, Postman, Swagger/OpenAPI, Storybook, VS Code, IntelliJ

Education

Malla Reddy College of Engineering and Technology August 2019 - June 2023
BTech in Computer Science and Engineering, CGPA: 8.38, Graduated: 8th June 2023

Achievements and Certificates

- MongoDB Dev Toolkit: Link
- Leetcode: (Highest Rating 1469 and solved 100+ problems).
- Participated in 100xDevs Solana Mini Hackathon.